Process Characterization Report

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCR-010

Process Development / Manufacturing Science & Technology (MSAT)

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

A280, absorbance at 280 nm; AMV, analytical method validation; BLA, Biologics License Application; CPP, critical process parameter; CQA, critical quality attribute; DNA, deoxyribonucleic acid; DS, drug substance; DV, diavolume; GPP, general process parameter; HCP, host cell protein; icIEF, imaged capillary isoelectric focusing; IgG1, immunoglobulin G subclass 1; KPP, key process parameter; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; PAR, proven acceptable range; QbD, quality by design; SD, standard deviation; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure; TFF, tangential-flow filtration; TMP, transmembrane pressure; UF/DF, ultrafiltration / diafiltration; WC-CPP, well-controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Ultrafiltration and diafiltration is the last unit operation of the A-Mab drug substance process. The step concentrates the pool from small-virus retentive filtration (Step 9), exchanges it into the formulation buffer by constant-volume diafiltration, and then adjusts it to the target drug substance concentration, which is the state in which the material passes to the drug product process. It forms no quality attribute and clears no impurity, so all 3 of its parameters are key process parameters and none of them is linked to a critical quality attribute. This report presents the characterization executed under PCP-010 on a qualified scale-down model (§3.1), within the risk scope set by RA-001.

The step delivered 54.6 kg of A-Mab at 75 g/L, which is the target concentration for the drug substance. Step yield was 97.1 %, and the cumulative yield of the drug substance train is 83.2 % of the mass produced in the production bioreactor (Figure 5.1). The loss across the step is attributed to product held in the membrane system and released only in part by the recovery flush. Buffer exchange is governed by the number of diavolumes, and under an ideal constant-volume model the set-point of 8 diavolumes leaves 0.034 % of a freely permeating feed species in the retentate, against 0.091 % at the low edge of the characterization range (§5.1).

Aggregate and charge variants were monitored across the operation, because the second concentration phase brings the product to the most concentrated state it reaches anywhere in the process. No formation or clearance of either attribute is attributed to this step. The aggregate content of the cation exchange pool is 0.73 % by SEC-HPLC against a drug substance limit of 5 %, and the simulated drug substance mean of 0.75 % is the same level within the batch-to-batch variation of the attribute (Table 7.1).

The campaign recorded 2 deviations at this step, and both were investigated and retained (§12). A transmembrane-pressure excursion reached 14.6 psi against a normal operating range top of 14 psi, and one run finished 3.5 g/L below the target concentration before being concentrated further and released. Both values sit inside the characterization ranges of the parameters concerned, and neither event changed a parameter classification or an operating range.

Three claims are deliberately not made for this step. It takes no viral clearance credit, it takes no impurity clearance credit, and it does not characterize the formulation. Formulation composition and the stability of A-Mab in it are established in the drug product programme, outside the drug substance transfer scope defined in PTP-001. What the step establishes is the delivery of the drug substance at its target concentration in the formulation buffer, carrying the product quality it received from the purification train (§5.3). The outcome rolls up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody-dependent cellular cytotoxicity. It is produced by fed-batch mammalian cell culture and purified on a platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration / diafiltration. The steps and their roles are listed in Table 1.1.

Table 1.1: The A-Mab drug substance process train and each step's principal role in the control strategy.

Step	Unit operation	Principal role
3	Production Bioreactor	Forms the glycan, charge-variant and aggregate CQAs (design-space step)
4	Harvest and Clarification	Primary recovery and clarification; forms no product-quality CQA
5	Protein A Chromatography	Capture; sets leached Protein A; principal HCP and DNA clearance
6	Low-pH Viral Inactivation	Low-pH hold; sets the cumulative XMuLV (enveloped) clearance
7	Cation Exchange Chromatography	Polish; principal aggregate reduction; major HCP/DNA/leached-PA clearance
8	Anion Exchange Chromatography	Flow-through polish; sets the cumulative MVM clearance; clears XMuLV/HCP/DNA
9	Small-Virus Retentive Filtration	Dedicated small-virus removal; principal MVM clearance
10	Ultrafiltration / Diafiltration	Formulation / mass balance; delivers drug substance at target concentration

Ultrafiltration and diafiltration is Step 10 and the last operation before the drug substance is filled. It is a tangential-flow filtration operation on a membrane whose nominal molecular weight cut-off retains the antibody and passes buffer species, and it runs in three phases. The pool from small-virus retentive filtration is first concentrated, then diafiltered at constant volume against the formulation buffer, and then concentrated a second time to the target drug substance concentration, after which a recovery flush displaces part of the retentate hold-up into the pool.

The step changes the concentration and the buffer of the product. It does not change the molecule. The membrane retains monomer and aggregated forms alike, so no size variant is separated, and no charge-based or affinity-based mechanism operates, so no charge variant and no glycosylation variant is separated either. The step accordingly forms no critical quality attribute and takes no clearance credit in the control strategy, and its parameters govern process performance only, which is the reason all of them are classified as key process parameters (§8).

1.2 Objectives

The objectives of the characterization were as follows.

- Confirm that the operation delivers the drug substance at its target concentration over the ranges assessed.
- Quantify step yield and close the product mass balance for the drug substance train.
- Establish that buffer exchange is complete at the low edge of the diavolume range.
- Confirm that the quality attributes carried into the step are preserved across it.
- Justify the classification of each parameter and the operating ranges that pass into the control strategy.

1.3 Regulatory and scientific basis

This report is a Stage 1 process design record in the sense of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), and its structure follows that guidance. The enhanced development approach of ICH Q8 and the quality risk management framework of ICH Q9 apply (International Council for Harmonisation 2009, 2023b), with criticality treated as a continuum and not as a binary label. ICH Q11 provides the development and manufacture framework for the drug substance (International Council for Harmonisation 2012). The A-Mab case study is the source of the process model, the quality attribute framework and the risk methodology used across the package (CMC Biotech Working Group 2009). No viral clearance is claimed for this step, so the viral safety framework of ICH Q5A is not applied here (International Council for Harmonisation 2023a). The modular clearance claims are made at Steps 6, 8 and 9 and are consolidated in PCMR-001.

1.4 Related documents

The package documents that scope, plan and consume this report are listed in Table 1.2.

Table 1.2: Cross-references to the sibling documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-010	Process Characterization Plan (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Ultrafiltration and diafiltration on this platform is established from three prior humanized IgG1 products (X-Mab, Y-Mab and Z-Mab) and from the A-Mab clinical and engineering campaigns. The membrane chemistry, the nominal cut-off and the three-phase operating strategy (concentrate, diafilter, concentrate) are common to all four products. What differs between products is the target concentration and the composition of the formulation buffer, and both of those are set by the drug product programme. The platform history is therefore transferable to A-Mab for the operation itself, and it is not transferable for the formulation, which is why this report characterizes the first and defers the second (§10).

The mechanistic expectation for each parameter was fixed before the study, and it is what sets the ranges assessed. Transmembrane pressure drives permeate flux, and it also drives concentration polarization at the membrane surface, so that above a product-specific pressure the flux stops rising with pressure and a gel layer forms, which lengthens the operation and raises the pressure needed to hold a given flux. The number of diavolumes sets the completeness of buffer exchange, and for a species that passes the membrane freely the residual fraction falls exponentially with the diavolume count. The final concentration sets the viscosity of the retentate, the mass recoverable from a fixed hold-up volume, and the degree of self-association available to the product.

These expectations account for the characterization ranges in Table 4.1. The diavolume range extends 1 diavolume above the normal operating range, so that the yield cost of over-processing could be observed, and its low edge is the platform value at which complete exchange has been demonstrated for the three prior products. The transmembrane pressure range extends 1 psi above the normal operating range, because a transient pressure rise during a concentration phase is the most common excursion at this step, and the deviation recorded during execution (DEV-010-01) was of exactly that kind. The concentration range extends on both sides of target, because the low side tests the recoverable mass and the high side tests the product in its most concentrated state.

2.2 Quality attributes in scope

No critical quality attribute is assigned to this step in the register. The attributes at risk from a concentration operation are the size variants, and, over the duration of the operation, the charge variants. Both were monitored across the step, and both are given in Table 2.1 with their acceptance criteria and criticality.

Table 2.1: Quality attributes monitored across the ultrafiltration / diafiltration step. Neither is formed nor cleared here; both are monitored for preservation.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

Aggregates are formed in the production bioreactor and reduced at cation exchange, which is the only aggregate polishing step in the train (PCR-003 and PCR-007). Acidic charge variants are formed in the production bioreactor and are not cleared downstream (PCR-003). This step is credited with neither the formation nor the clearance of either attribute, and it is monitored against both so that a change would be detected if one occurred.

Host cell protein, residual DNA and leached Protein A pass through the step. Their levels in the drug substance are the levels delivered by the chromatographic steps, so the credit for them belongs to PCR-005, PCR-007 and PCR-008, and the viral clearance claims belong to PCR-006, PCR-008 and PCR-009. All of these contributions are consolidated in PCMR-001, which is the document that states the drug substance position for each attribute.

2.3 Risk-based prioritization and parameter selection

RA-001 assigned every parameter of this step to univariate assessment, and no multivariate design was planned or executed. The assignment follows from the severity axis of the risk ranking. A parameter whose failure mode cannot reach a critical quality attribute scores at the bottom of the severity scale, and the combined score then falls below the threshold at which a multivariate design is required. The three failure modes at this step are a low step yield, an incomplete buffer exchange and a concentration away from target, and each of them is detected in process, is correctable within the batch, and leaves the molecule itself unchanged.

One consequence of that assignment is carried through this report. No interaction between the three parameters has been quantified, so the ranges in §6 are single-parameter ranges and not a multivariate region. How that limitation is managed is stated in §10.

3 Materials and methods

3.1 Scale-down model and its qualification

The scale-down model is a tangential-flow filtration system operated at the same membrane loading as the commercial skid, expressed as grams of protein per square metre of membrane area. Membrane chemistry, nominal cut-off and channel geometry are the same at both scales. Crossflow is set to match channel shear, transmembrane pressure is matched directly, and the diavolume count is dimensionless and transfers between scales unchanged. Qualification compared step yield and final concentration on the model against the commercial skid operated at set-point conditions, under SOP-1001 and SOP-2013.

The model has one known limitation, and it bears on the yield claim and not on the concentration and exchange claims. Hold-up volume does not scale in proportion to membrane area, and hold-up is the dominant term in the step loss, so the model reproduces the commercial recovery less faithfully than it reproduces the other two outcomes. The consequence is stated in §10 and is managed by confirmation at commercial scale in Stage 2.

The scale-down system used for the characterization was the TFF skid EQ-TFF-142. It was within its calibration interval throughout execution, with calibration next due on 2026-04-22.

3.2 Operation

Operation follows SOP-2013. The feed is the filtrate pool from small-virus retentive filtration, which is transferred to the retentate vessel and recirculated across the membrane at the set-point crossflow. The first concentration phase raises the protein concentration to the diafiltration concentration. Constant-volume diafiltration then replaces the feed buffer with the formulation buffer, with permeate removal balanced by addition of formulation buffer, until the set diavolume count is reached. A second concentration phase brings the retentate to the target drug substance concentration, and a recovery flush of formulation buffer displaces part of the hold-up volume into the pool. The pooled retentate and flush are the drug substance.

Inputs to the step are controlled to specification and were not characterized. The formulation buffer is a material input, prepared to the composition fixed by the drug product programme and released against its own specification, so its identity and quality are controlled but its composition is not a subject of this study. The membrane is qualified by lot and by normalized water permeability before use, and its reuse is controlled under the life-cycle provisions of SOP-2013.

3.3 Analytical methods

The procedures and validated methods used at this step are listed in Table 12.3 in Appendix B. Protein concentration is measured by absorbance at 280 nm (AMV-3019), and that assay is used in process at the end of each concentration phase and again on the final pool, where it is the measurement on which the batch is dispositioned. Size variants are measured by SEC-HPLC (AMV-3011) and charge variants by imaged capillary isoelectric focusing (AMV-3013). The intermediate precision of the concentration assay is 1.2 % relative standard deviation, which is 0.9 g/L at the target concentration, and its limit of quantitation is 0.5 g/L. That precision is the reference against which the concentration deviation in §12 is judged.

3.4 Sampling plan

Samples were taken from the feed to the step, from the retentate at the end of diafiltration, and from the final pool after the recovery flush. Concentration was measured on every sample. Size variants and charge variants were measured on the feed and on the final pool, because the feed-to-pool comparison is the measurement that answers whether the operation changes the product. Buffer exchange is followed in process by conductivity and pH of the retentate at the end of diafiltration, which is the routine batch control under SOP-2013.

3.5 Statistical methods

No statistical model was fitted for this step. Each parameter was assessed over its characterization range with the other two held at their set-points, so the data support a range statement per parameter and not a response surface, which is why §6 reports proven acceptable ranges instead of a design space.

Commercial-scale capability is estimated by Monte-Carlo simulation of 2,000 batches with every process parameter varying inside its normal operating range, and is reported as a one-sided Cpk against the acceptance criterion for each attribute. Because this step neither forms nor clears an attribute, its contribution to capability is the preservation of the values delivered to it, so the capability figures in §7 are drug substance figures whose credit lies with the steps named there.

4 Study design

The design is univariate and it follows RA-001 and PCP-010. Each parameter was assessed at the edges of its characterization range with the other two at their set-points, and verification runs were executed at the set-point condition. The parameters, their set-points, their normal operating ranges, their characterization ranges and their final classification are given in Table 4.1.

Table 4.1: Parameters of the ultrafiltration / diafiltration step, with set-points, normal operating ranges, characterization ranges, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Number of diavolumes	DV	8	7–9	7–10	KPP	univariate
Transmembrane pressure	psi	12	10–14	10–15	KPP	univariate
Final DS concentration	g/L	75	70–80	65–85	KPP	univariate

Every characterization range is wider than the corresponding normal operating range on at least one side, and that margin is what the range claims in §6 rest on. Diavolumes were assessed from 7 to 10, where the low edge is the condition at which buffer exchange is least complete and is therefore the worst case for the exchange claim, and the high edge tests whether extended processing costs product. Transmembrane pressure was assessed to 15 psi. The final concentration was assessed from 65 to 85 g/L against a normal operating range of 70 to 80 g/L.

The responses recorded for each condition were the step yield, the final concentration by A280, the conductivity and pH of the retentate at the end of diafiltration, and the size and charge variant profiles of the final pool (§3.4). No screening design and no response-surface design was run at this step, and none was required by RA-001.

5 Results

5.1 Concentration and buffer exchange

The operation delivered the drug substance at 75 g/L, which is the target concentration and the centre of the normal operating range of 70 to 80 g/L. Concentration is measured by A280 at the end of the final concentration phase, and the batch is adjusted before the pool is released. One run finished below target and is described in §12.

Buffer exchange is complete at the low edge of the diavolume range. Under an ideal constant-volume diafiltration, in which the retentate is fully mixed and the species passes the membrane freely, the residual fraction of a feed buffer species is the exponential of the negative diavolume count. At the set-point of 8 diavolumes that fraction is 0.034 % of the starting concentration, at 7 diavolumes it is 0.091 %, and at 10 diavolumes it is 0.0045 %.

Two bounds apply to that calculation, and neither of them is removed by the data. The first is that the calculation is the design basis for the diavolume set-point and is not the batch acceptance test, which is the conductivity and pH of the retentate at the end of diafiltration (SOP-2013). The second is that it applies only to species that pass the membrane freely, because a species that associates with the product is not removed by diafiltration at all, and clearance of that kind is credited to the chromatographic steps (PCR-005, PCR-007 and PCR-008) and not to this one.

5.2 Step yield and the process mass balance

The step recovered 54.6 kg of the 56.2 kg delivered to it, which is a step yield of 97.1 % and a loss of 1.6 kg, and the recovered mass corresponds to 728 L of drug substance at the target concentration. The loss is attributed to product held in the membrane system and the transfer lines after the recovery flush, and that attribution is supported by the investigation of DEV-010-02, whose root cause was residual hold-up volume (§12). At 2.9 %, the loss is the second largest of the 7 recovery and purification steps in the train.

This is the last step of the drug substance train, so the mass balance closes here. Cumulative yield from the production bioreactor to the formulated drug substance is 83.2 %, from a nominal batch of 15,000 L of culture, and the delivered mass is 54.6 kg. The step-by-step profile is shown in Figure 5.1 and tabulated in Table 12.2 in Appendix A. The largest single step loss anywhere in the train is 4.8 %, so the recovery burden is spread across the recovery and chromatographic steps instead of being concentrated in any one of them.

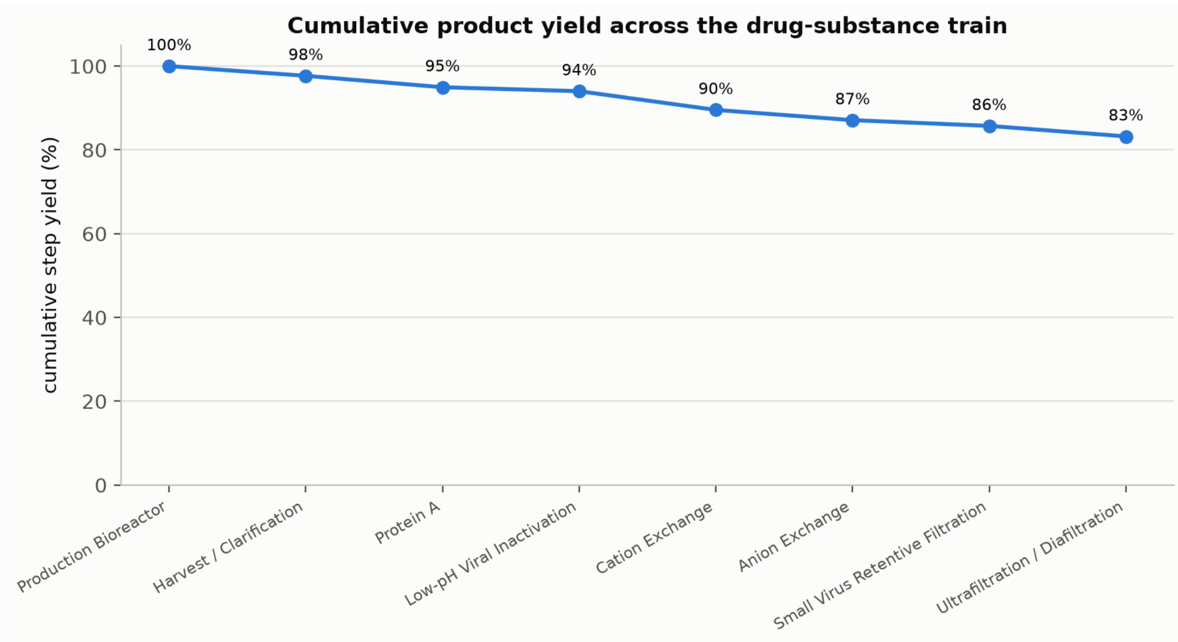

Figure 5.1: Cumulative product yield across the A-Mab drug substance train, from the production bioreactor to the formulated drug substance.

5.3 Product quality across the operation

The quality attributes of the drug substance are the attributes delivered to this step. The aggregate content of the cation exchange pool is 0.73 % of the total by SEC-HPLC, against a drug substance acceptance limit of 5 %, and no formation or clearance of aggregate is attributed to anion exchange, to small-virus retentive filtration or to this step. The simulated drug substance mean is 0.75 % (Table 7.1), which is the cation exchange pool level within the batch-to-batch variation of the attribute. Acidic charge variants behave the same way, and their acceptance interval of 18 to 40 % is met at drug substance.

Concentration to 75 g/L is the most concentrated state the product reaches in the process, and it is the condition under which self-association is most likely. The comparison of feed and final pool by SEC-HPLC and icIEF is the evidence that the operation preserves what it receives, and it is the reason both assays sit in the sampling plan (§3.4) and in the control strategy (§9). That comparison covers the concentrations assessed in this study and does not extend above 85 g/L.

Residual DNA in the drug substance is 0.0001 ng per dose against a limit of 0.001 ng per dose. Host cell protein, leached Protein A and residual DNA are cleared upstream and are carried across this step unchanged, so their capability is reported in PCR-005, PCR-007 and PCR-008, and the drug substance position for each is consolidated in PCMR-001.

6 Operating ranges and proven acceptable ranges

This step has no design space. The characterization was univariate, so what the data support is a proven acceptable range for each parameter and not a multivariate region in three parameters, and this section accordingly replaces the design-space section used for the characterized steps. The substitution follows from the risk-based study assignment in RA-001 (§2.3), and it does not follow from an incomplete study.

The proven acceptable range of each parameter is its characterization range in Table 4.1, and the normal operating range sits inside it in every case. Diavolumes are proven acceptable from 7 to 10, and the binding condition is the low edge, at which buffer exchange is least complete (§5.1). Transmembrane pressure is proven acceptable from 10 to 15 psi, and the excursion recorded in DEV-010-01 fell inside that range. Final concentration is proven acceptable from 65 to 85 g/L, which is the range over which the size and charge variant profiles of the pool were compared with those of the feed (§5.3).

The worst-case condition for the step is the low edge of the diavolume range combined with the high edge of the concentration range, because that combination gives the least complete buffer exchange at the highest retentate viscosity. It was not executed as a single run, since the study was univariate, so it is identified here and it is not claimed as characterized.

Three bounds apply to these ranges. They hold over the values assessed and on the scale-down model described in §3.1, and they do not extend beyond the characterization ranges. They are single-parameter ranges, so simultaneous movement of two parameters to opposite edges is not covered by them. Because there is no design space for this step, the ICH Q8 provision that movement within a design space is not a change does not apply here (International Council for Harmonisation 2009): movement inside the normal operating range is routine operation, and movement outside it but inside the proven acceptable range is assessed under the pharmaceutical quality system before execution (International Council for Harmonisation 2008).

7 Process capability and robustness

The drug substance meets every acceptance criterion at commercial scale, and none of that capability is earned at this step. Capability was estimated by Monte-Carlo simulation of 2,000 batches with all process parameters varying inside their normal operating ranges. The two attributes monitored across this operation are given in Table 7.1.

Table 7.1: Commercial-scale capability of the attributes monitored across the ultra-filtration / diafiltration step, from a Monte-Carlo simulation of the whole process.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26

Aggregates carry a one-sided Cpk of 16.1 against the upper limit of 5 %, with a simulated mean of 0.75 %, and the margin is that large because the aggregate limit sits well above the level the cation exchange step delivers (PCR-007). Acidic charge variants carry a Cpk of 4.26, which is the tighter of the two, and that figure is a property of the attribute's acceptance interval and of the variation introduced upstream at the production bioreactor (PCR-003). Neither figure is evidence about this step beyond the absence of a contribution from it.

The tightest capability in the drug substance is neither of these. The minimum one-sided Cpk across all attributes is 1.51, and it belongs to the cumulative parvovirus clearance, which is credited to anion exchange and small-virus retentive filtration (PCR-008 and PCR-009). The consolidated capability position is given in PCMR-001.

Two bounds apply to the figures above. They assume every parameter inside its normal operating range, and they rest on the scale-down models qualified for each step of the train, so they are model estimates and are confirmed at commercial scale during Stage 2.

8 Parameter classification

All 3 parameters of the step are key process parameters. None is critical and none is well-controlled critical, because no failure mode at this step reaches a critical quality attribute. The reason for each classification is given below, and the classification rule is the one in SOP-4001.

- The number of diavolumes is a key process parameter, because it sets the completeness of buffer exchange, which is a performance attribute of the formulated pool and not a property of the molecule. It is fixed in the batch record and counted automatically during the phase, so the risk of operating outside its range is low.
- Transmembrane pressure is a key process parameter. It governs flux, and therefore the duration of the operation, and at the top of its range it also governs the risk of a fouling transient of the kind recorded in DEV-010-01, but it is monitored continuously and any excursion is correctable within the batch.
- The final drug substance concentration is a key process parameter, and it is the deliverable of the step, so a value away from target is a process performance failure and not a quality failure. It is verified by A280 before the pool is released, and a low result is corrected by further concentration, as DEV-010-02 shows.

This step contributes 3 of the 10 key process parameters in the drug substance process, and none of the 21 quality-linked parameters (CPP and WC-CPP). The classification counts across the whole train are consolidated in PCMR-001.

9 Contribution to the control strategy

The step contributes four elements to the control strategy for the drug substance. The three parameters are held inside the normal operating ranges in Table 4.1, and the set-points are carried in the batch record. Buffer exchange is verified in process by the conductivity and pH of the retentate at the end of diafiltration. The concentration of the final pool is verified by A280 (AMV-3019) before the pool is released, with the provision for further concentration that DEV-010-02 exercised. Product quality across the step is verified by SEC-HPLC and icIEF on the feed and on the final pool.

Two life-cycle controls attach to the step and not to a single batch. Membrane reuse is controlled by normalized water permeability and by a maximum cycle count under SOP-2013, and the formulation buffer is a controlled material input released against the specification set by the drug product programme.

What the step does not control matters as much to the assessor as what it does. It clears no impurity and no clearance credit is taken for it, so the impurity levels in the drug substance are those established at Steps 5, 7 and 8, and it makes no viral clearance claim, so the modular claim rests entirely on Steps 6, 8 and 9. It does not establish the formulation and it does not establish the stability of the drug substance in that formulation. Those studies are reported with the drug product and are outside the transfer scope in PTP-001.

10 Discussion

Two limitations bound what this report establishes, and the first is the absence of a multivariate design. Only single-parameter ranges are supported, so a simultaneous excursion in two parameters is not covered by the data and is handled under change control. The residual risk is small, because the three parameters act on different outcomes: diavolumes act on exchange completeness, transmembrane pressure acts on flux, and concentration acts on recoverable mass. The risk is not zero, because all three act on the duration of the operation and therefore on the time the product spends at high concentration, and that coupling is the reason the worst-case condition in §6 is identified even though it was not executed as a run.

The second limitation is the scale-down model. Hold-up volume does not scale in proportion with membrane area, and hold-up is the dominant term in the step loss, so the model predicts commercial step yield with less confidence than it predicts concentration and buffer exchange. The step yield reported in §5.2 is the model value, and it is confirmed against the commercial skid during Stage 2. The concentration and exchange claims do not depend on hold-up and are unaffected by this limitation.

The 2 recorded deviations were transient and both were retained. The pressure excursion reached 14.6 psi, which is 0.6 psi above the normal operating range and 0.4 psi below the top of the characterization range, and the concentration shortfall was 3.5 g/L on one run and was corrected before release. Neither event changed a classification or a range, and both are described in §12.

Formulation characterization is deferred and not omitted. The composition of the formulation buffer, the stability of A-Mab in it, and the drug substance hold, freeze and thaw conditions are established in the drug product development programme, and that programme is where an assessor reading for the formulation argument should look. This report covers the delivery of the drug substance into that buffer at the target concentration, which is where the drug substance transfer scope defined in PTP-001 ends.

No parameter at this step required the most stringent class of control. That outcome follows from the mechanism of the operation and not from a wide operating margin, and it would not change if the ranges were narrowed, because a step that neither forms nor clears a quality attribute cannot hold a critical process parameter under the classification rule in SOP-4001.

11 Conclusions

The ultrafiltration and diafiltration step is well understood and its operation is under control. It delivered 54.6 kg of A-Mab at 75 g/L in the formulation buffer, at a step yield of 97.1 % and a cumulative train yield of 83.2 %. Buffer exchange is complete at the low edge of the diavolume range. The quality attributes carried into the step are preserved across it, and the drug substance meets every acceptance criterion at commercial scale, with a minimum one-sided Cpk of 1.51 that belongs to the cumulative parvovirus clearance credited to Steps 8 and 9.

All 3 parameters are classified as key process parameters, with the proven acceptable ranges in Table 4.1. The 2 recorded deviations were investigated and retained, and neither altered a classification or a range. No design space, no clearance credit and no viral clearance claim is made for this step, and the formulation itself is characterized in the drug product programme. These conclusions hold over the ranges assessed and on the qualified scale-down model, and they roll up into PCMR-001.

12 Deviations from the plan

Execution of PCP-010 recorded 2 deviations. Both were investigated, impact assessed and retained, and neither changed a parameter classification or an operating range. The register is given in Table 12.1.

Table 12.1: Register of deviations recorded during execution of the ultrafiltration / diafiltration characterization.

Deviation	Summary	Detected during	Disposition
DEV-010-01	Transmembrane-pressure excursion above NOR during concentration	in-process TMP monitoring	retained
DEV-010-02	Final DS concentration below target on one run; topped up	final concentration check	retained

12.1 DEV-010-01 Transmembrane-pressure excursion during concentration

Transmembrane pressure reached 14.6 psi during a concentration phase, which is 0.6 psi above the top of the normal operating range of 14 psi. The excursion was detected by in-process monitoring on the TFF skid (EQ-TFF-142), after which crossflow was reduced and the phase was completed at the set-point pressure.

The investigation attributed the excursion to a transient fouling of the membrane during concentration, which raised the pressure needed to hold the set flux. The membrane passed its normalized water permeability check after cleaning, and that result is consistent with a transient and reversible event and not with membrane damage.

The value reached is inside the characterization range of the parameter, whose upper edge is 15 psi, so the run was executed inside the conditions the study covers, with a margin of 0.4 psi to the top of that range. The run was retained and its data were used as recorded. The event is one of the reasons transmembrane pressure is monitored continuously in the control strategy (§9).

12.2 DEV-010-02 Final concentration below target on one run

The final concentration of one run was 71.5 g/L, which is 3.5 g/L below the target of 75 g/L, or 4.7 % low. The value is inside the normal operating range, whose lower edge is 70 g/L, so no material was outside a studied condition at any point. The shortfall is 3.9 times the intermediate precision of the concentration assay, so it is a real shortfall and not an analytical artefact.

The investigation attributed the shortfall to residual hold-up volume in the membrane system, which retained product that the recovery flush did not displace. The pool was concentrated further to the target and re-assayed by A280 (AMV-3019) before release, and the batch was dispositioned as conforming.

The run was retained. This event is the basis for the in-process concentration check in §9, and it is the observation that supports the statement in §10 that the scale-down model predicts step yield with less confidence than it predicts concentration.

Appendix A — Drug substance mass balance across the process train

The step and cumulative yields of every unit operation in the drug substance train are given in Table 12.2. The values are those of the seeded process model and are the source of the figures quoted in §5.2.

Table 12.2: Step and cumulative product yield for each unit operation of the A-Mab drug substance process.

Step	Unit operation	Step yield (%)	Cumulative yield (%)
3	Production Bioreactor	100	100
4	Harvest / Clarification	97.7	97.7
5	Protein A Chromatography	97.2	95
6	Low-pH Viral Inactivation	99	94
7	Cation Exchange Chromatography	95.2	89.5
8	Anion Exchange Chromatography	97.3	87.1
9	Small Virus Retentive Filtration	98.4	85.7
10	Ultrafiltration / Diafiltration (formulation)	97.1	83.2

Appendix B — Analytical methods summary

The controlled procedures and validated analytical methods referenced by this report are listed in Table 12.3. Method performance for the concentration assay is quoted in §3.3; full validation summaries are held in the referenced validation reports.

Table 12.3: Standard operating procedures and validated analytical methods for the ultrafiltration / diafiltration step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3019	Protein Concentration by A280 (UV)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

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